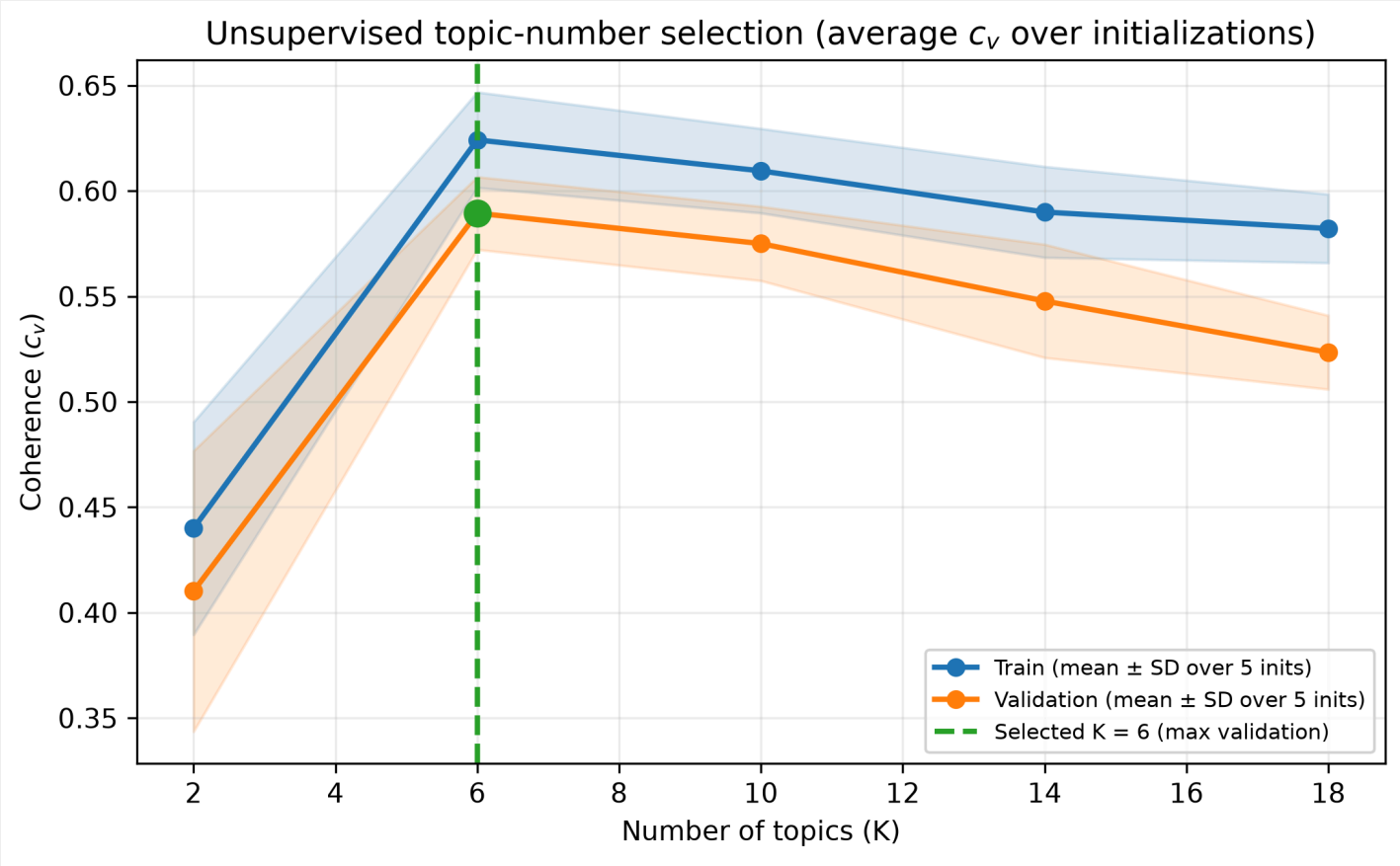
LDA topic-number selection and stability report

Similarity metric throughout: **cosine** between aligned `P(day-type | topic)` distributions (Hungarian matching to the published reference).

---

1. Unsupervised selection of K

We compared several candidate values of **K** using mean topic coherence (**c_v**) over 5 initializations (train/validation 70/30).
Rule: maximise mean validation coherence → **K = 6**.



Published LDA: K = 6 re-trained with many passes (~100 000), `models/lda/LDA_model_vdec25.gensim`.

---

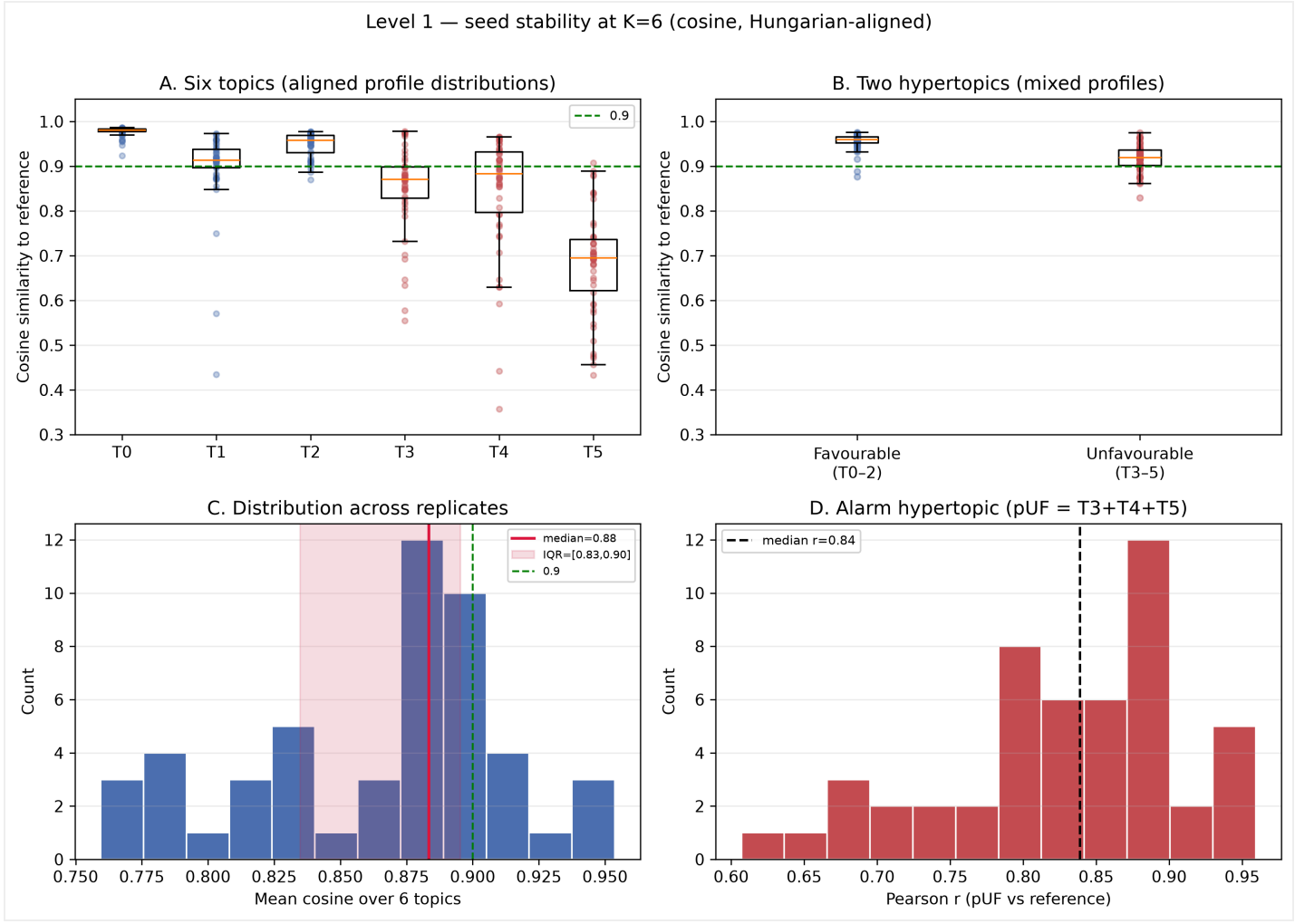
2. Level 1 — Stability to random seed

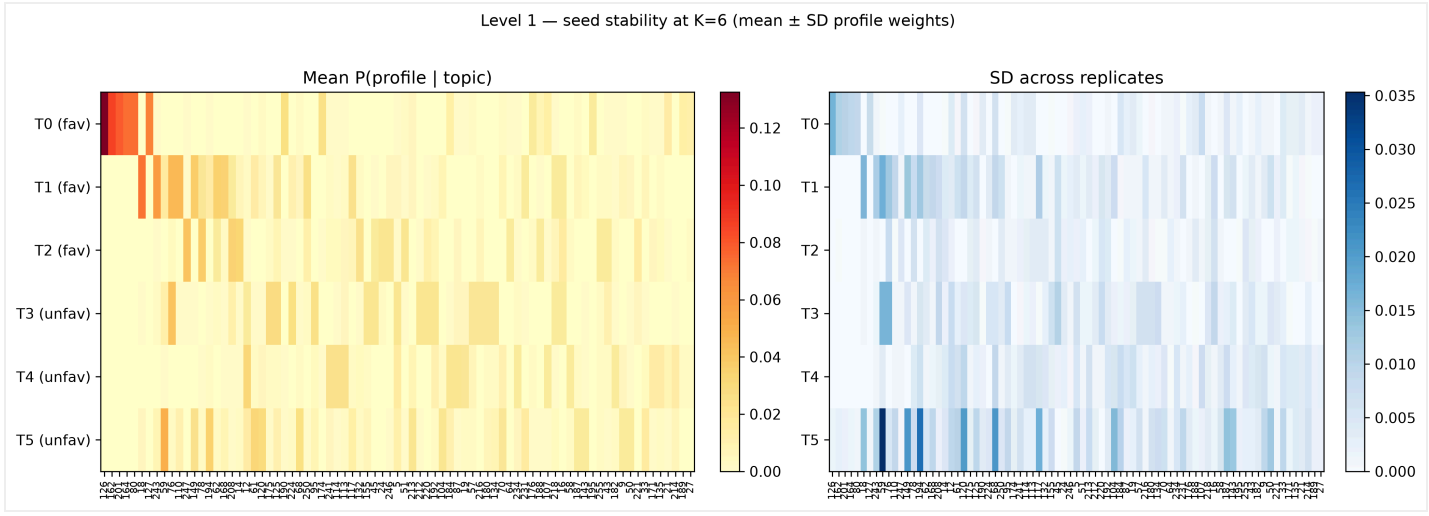
Re-fit LDA **50 times** at K = 6 (all patients; different seeds). Align topics to the reference; report **cosine** similarity of topic–profile distributions and patient **pUF** concordance.

Headline (cosine)

Summary	Value
Median cosine (all topic–replicate pairs)	**0.912**
IQR	**[0.836, 0.964]**
Median of per-replicate mean cosine (6 topics)	**0.883**
IQR of per-replicate mean cosine	**[0.835, 0.895]**
Favourable hypertopic cosine (median)	**0.960**
Unfavourable hypertopic cosine (median)	**0.919**
Patient pUF Pearson r (median [IQR])	**0.839 [0.798, 0.887]**

Topic	Hypertopic	Cosine median [IQR]
T0	favourable	0.981 [0.977, 0.983]
T1	favourable	0.914 [0.897, 0.938]
T2	favourable	0.958 [0.930, 0.969]
T3	unfavourable	0.871 [0.829, 0.898]
T4	unfavourable	0.883 [0.796, 0.932]
T5	unfavourable	0.695 [0.622, 0.736]





---

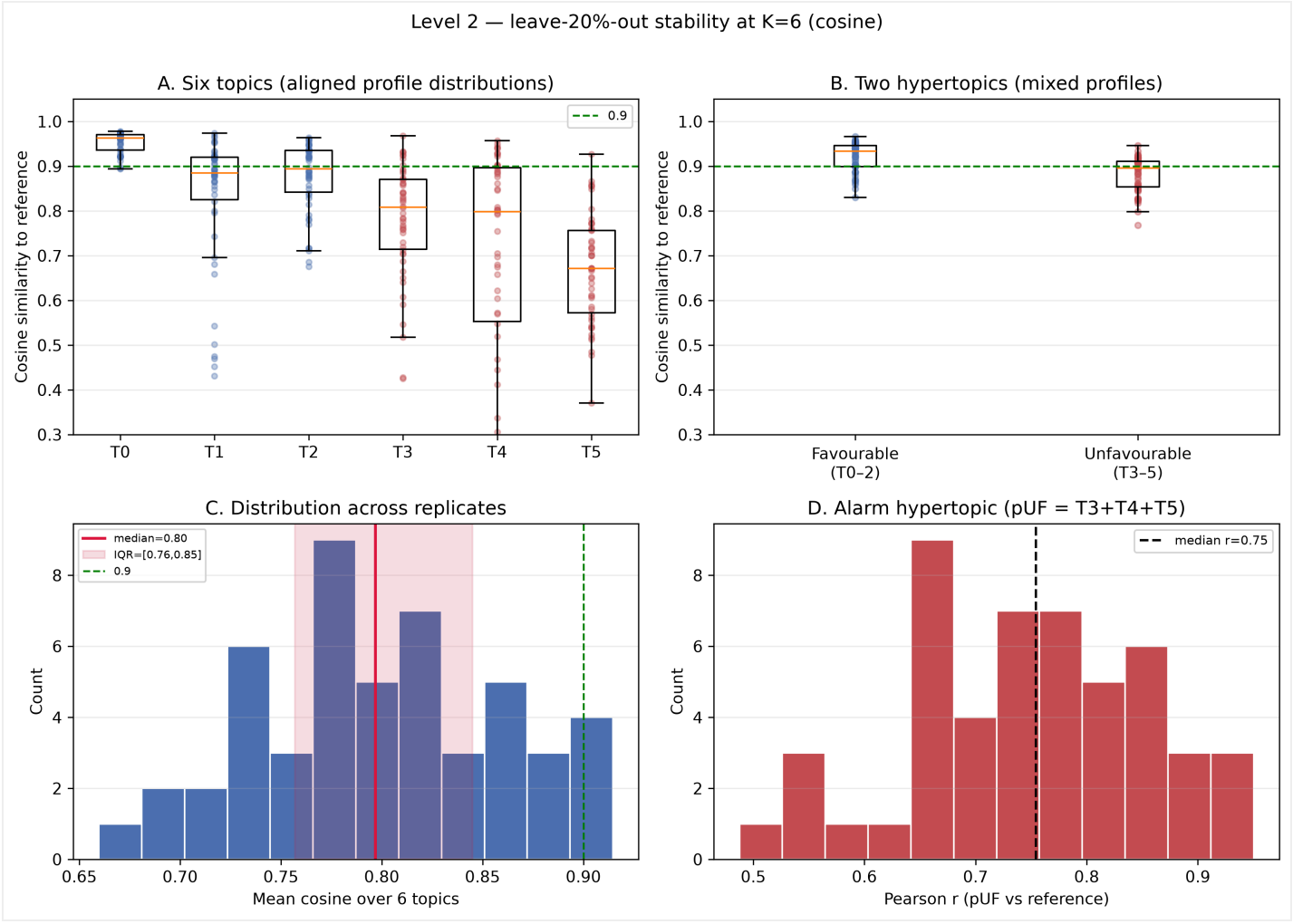
3. Level 2 — Stability to data (leave-20%-out)

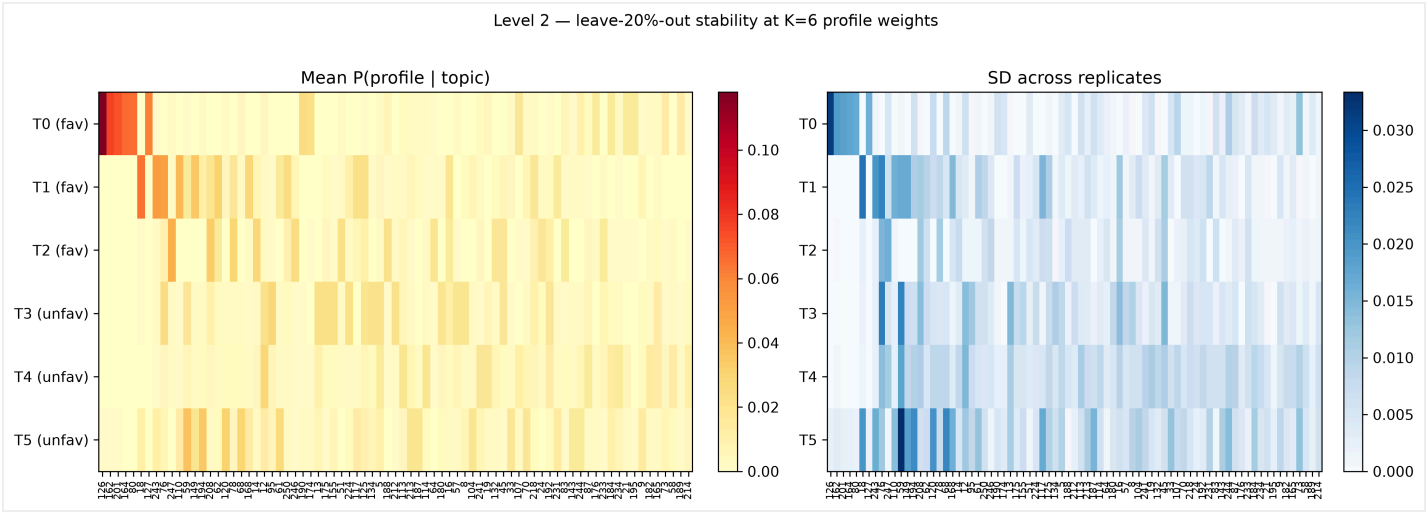
Same K = 6 and cosine alignment, but each replicate **drops 20% of patients at random** (n = 50 replicates) and re-fits LDA on the remainder.

Headline (cosine)

Summary	Value
Median cosine (all topic–replicate pairs)	**0.867**
IQR	**[0.706, 0.927]**
Median of per-replicate mean cosine	**0.797**
Favourable / unfavourable hypertopic cosine (median)	**0.933** / **0.896**
Patient pUF Pearson r (median [IQR])	**0.754 [0.660, 0.830]**

Topic	Hypertopic	Cosine median [IQR]
T0	favourable	0.963 [0.936, 0.970]
T1	favourable	0.885 [0.825, 0.920]
T2	favourable	0.894 [0.842, 0.936]
T3	unfavourable	0.808 [0.714, 0.870]
T4	unfavourable	0.798 [0.553, 0.896]
T5	unfavourable	0.671 [0.572, 0.757]





---

4. Interpretation

- **K = 6** maximises mean validation coherence among the candidates evaluated.

- **Level 1 (seed):** topic–profile cosine has **median > 0.9**; hypertopics and **pUF** remain highly concordant → the alarm input is stable to re-initialisation.
- **Level 2 (leave-20%-out):** cosine remains high for most topics/hypertopics under patient subsampling; use the heatmaps (mean ± SD of day-type weights) to show which profiles stay dominant.
- Alarm analyses are **not** re-run unless pUF concordance were poor.

---

5. Files

Path	Content
`coherence_coarse_grid.csv`	K-selection
`level1_seed_*.csv` / `level1_seed_summary.json`	Seed stability (cosine)
`level2_resample_*.csv` / `level2_resample_summary.json`	Leave-20%-out (cosine)
`../figures/lda_coherence_k_selection.*`	K figure
`../figures/stability/lda_stability_level1_cosine.*`	Level 1
`../figures/stability/lda_stability_level2_resample_cosine.*`	Level 2
`../figures/stability/*_profile_heatmap.*`	Mean ± SD profile weights

```
.venv/bin/python scripts/03_analysis/lda/run_lda_stability.py --mode both --n-reps 50 --p
```

Generated 2026-09-30